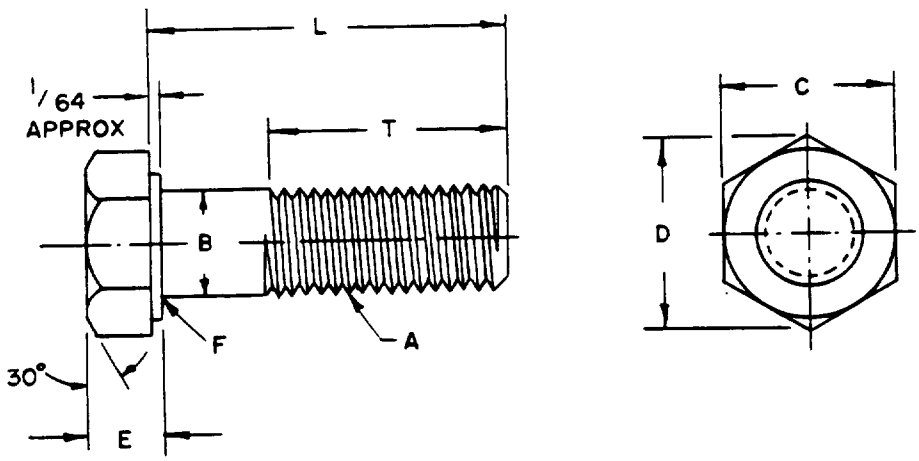
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FED. SUP. CLASS 5306



APPROVED 8 MAY 1957 REVISED (A) 19 AUGUST 1966

(Project 5306-N011Sh) (A) For changes see sheets 1 through 4

CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS 16206 (SHIPS)
Ships	A - N - AF -	BOLTS, MACHINE, HEXAGON HEAD, REGULAR SEMI-FINISHED, ALUMINUM ALLOY, UNC-2A	
Procurement Specification (A) MIL-B-857		NONMAGNETIC	SHEET 1 OF 4

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PROCUREMENT SPECIFICATION (A) MIL-B-827

CUSTODIANS: Ships

OTHER INT. A- N- AF-

SUPERSEDES:

MILITARY STANDARD

BOLTS, MACHINE, HEXAGON HEAD, REGULAR SEMI-FINISHED, ALUMINUM ALLOY, UNC-2A, NONMAGNETIC

MS16206 (SHIPS)

SHEET 3 OF 4

Nominal size			1/4		5/16		3/8		7/16	
(A) Threads per inch			20-UNC2A		18-UNC2A		16-UNC2A		14-UNC2A	
(B) Body diameter	Max.		0.250		0.312		0.375		0.437	
	Min.		.217		.276		.334		.394	
(C) Width across flats	Max.		.437		.500		.562		.615	
	Min.		.425		.484		.544		.603	
(D) Width across corners	Max.		.505		.577		.650		.722	
	Min.		.484		.552		.620		.687	
(E) Head height	Max.		.163		.211		.243		.291	
	Min.		.150		.195		.226		.272	
(F) Radius	Max.		.031		.031		.031		.031	
	Min.		.009		.009		.009		.009	

(L) Length	Tolerance Size 1/4 - 3/8	Tolerance Size 7/16 - 1/2	Tolerance Size 5/8 - 1-1/4	MS part No.	FIIN	MS part No.	FIIN	MS part No.	FIIN	MS part No.	FIIN
1/4	±1/32	±1/16	±1/8	1		--		--		--	
3/8				2		25		45		71	
1/2				3		26		46		72	
5/8				4		27		47		73	
3/4				5		28		48		74	
1				6		29		49		75	
1-1/4				7		30		50		76	
1-1/2				8		31		51		77	
1-3/4				9		32		52		78	
2				10		33		53		79	
2-1/4				11		34		54		80	
2-1/2				12		35		55		81	
2-3/4				13		36		56		82	
3				14		37		57		83	
3-1/4				15		38		58		84	
3-1/2				16		39		59		85	
3-3/4				17		40		60		86	
4				18		41		61		87	
4-1/4				19		--		62		88	
4-1/2				20		--		63		89	
4-3/4				21		--		64		90	
5				22		--		65		91	
5-1/4				--		--		66		92	
5-1/2				--		--		67		93	
6				--		--		--		--	
6-1/2	±1/16	±3/32	±3/16	--		--		--		--	
7				--		--		--		--	
7-1/2				--		--		--		--	
8				--		--		--		--	
9				--		--		--		--	
10				--		--		--		--	
11				--		--		--		--	
12				--		--		--		--	

APPROVED 8 MAY 1957 REVISED (A) For changes see sheets 1 through 4

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PROCUREMENT SPECIFICATION (X) MIL-B-857

CUSTODIANS: Ships

OTHER INT. A: - N: - AF: -

SUPERSEDES:

MILITARY STANDARD

BOLTS, MACHINE, HEXAGON HEAD, REGULAR SEMI-FINISHED, ALUMINUM ALLOY, UNC-2A, NONMAGNETIC

MS16206

(SHIPS)

SHEET 3 OF 4

			1/2		5/8		3/4	
	Nominal size							
(A)	Threads per inch		13-UNC2A		11-UNC2A		10-UNC2A	
(B)	Body diameter	Max.	0.500		0.625		0.750	
		Min.	.450		.566		.685	
(C)	Width across flats	Max.	.750		.937		1.125	
		Min.	.725		.906		1.088	
(D)	Width across corners	Max.	.866		1.083		1.299	
		Min.	.826		1.033		1.240	
(E)	Head height	Max.	.323		0.403		0.493	
		Min.	.302		.378		.455	
(F)	Radius	Max.	.031		.062		.062	
		Min.	.009		.021		.021	

(L) Length	Tolerance			MS part No.	FIIN	MS part No.	FIIN	MS part No.	FIIN
	Size 1/4 - 3/8	Size 7/16 - 1/2	Size 5/8 - 1-1/4						
1/4	±1/32	±1/16	±1/8	---		---		---	
3/8				---		---		---	
1/2				96		---		---	
5/8				97		---		---	
3/4				98		126		155	
1				99		127		156	
1-1/4				100		128		157	
1-1/2				101		129		158	
1-3/4				102		130		159	
2				103		131		160	
2-1/4				104		132		161	
2-1/2				105		133		162	
2-3/4				106		134		163	
3				107		135		164	
3-1/4				108		136		165	
3-1/2				109		137		166	
3-3/4				110		138		167	
4				111		139		168	
4-1/4				112		140		169	
4-1/2				113		141		170	
4-3/4				114		142		171	
5				115		143		172	
5-1/2				116		144		173	
6				117		145		174	
6-1/2	±1/16	±3/32	±3/16	118		146		175	
7				119		147		176	
7-1/2				120		148		177	
8				121		149		178	
9				122		---		179	
10				123		---		180	
11				---		---		---	
12				---		---		---	

FED. SUP. CLASS 5306

APPROVED 8 MAY 1957

REVISED (A) For changes see sheets 1 through 4

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FED. SUP. CLASS 5306

(A) Material: Aluminum alloy shall be 6061 of QQ-A-200/8.
(A) Protective coating: Anodize in accordance with MIL-C-5541 or MIL-A-8625.
Threads: Threads shall be UNC-2A in accordance with Handbook H28.
Thread length: Minimum thread length shall be twice the nominal size (diameter) plus 1/4 inch for bolt lengths 6 inches and less, and twice the nominal size plus 1/2 inch for bolt lengths over 6 inches. Bolts too short for minimum thread length shall be threaded as close to the head as practicable.
Dimensions: All dimensions are in inches, unless otherwise specified.

Notes:
1. In the case of conflict with any document referenced herein, this standard shall govern.
2. Referenced documents shall be the issue in effect on date of invitation for bids.

APPROVED 8 MAY 1957 REVISED (A) For changes see sheets 1 through 4

CUSTODIANS	OTHER INT.	MILITARY STANDARD	MS16206 (SHIPS)
Ships	A - N - AF -	BOLTS, MACHINE, HEXAGON HEAD, REGULAR SEMI-FINISHED, ALUMINUM ALLOY, UNC-2A	
Procurement Specification (A) MIL-B-857		NONMAGNETIC	SHEET 4 OF 4

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